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# Buckling Analysis of Cylindrical Shells using Stochastic Finite Element Method with Random Geometric Imperfections

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**Abstract**

Thin-walled cylindrical shells often exhibit buckling failure and the experimental buckling load is usually lower than calculation results from classical theory and simulation without geometrical imperfection. Besides, test results with carefully conducted similar specimens still have substantial scatter due to imperfection sensitivity. The nonlinear analysis with FEM can obtain a high-precision result comparing the experiment. However, this is almost impossible in practical engineering. The initial geometric imperfections of cylindrical shells are complex and random properties. Theoretically, these geometric imperfections can be described using a random field or a Fourier series representation. After that a random sample of the buckling bearing capacity of the cylindrical shell can be obtained using the stochastic finite element method within the subsequent analysis. This paper presents the investigation of buckling analysis of cylindrical shells under axial compression considering the randomness of initial geometric imperfections, and the stochastic FEM is employed to calculate the statistical sample of the shell-buckling load based on a series of random fields and Fourier series representation of geometric imperfections.

**Keywords**

Cylindrical shell, Buckling, Stochastic Finite Element Method, Random field, Geometric imperfection, Fourier series representation, Maximum Entropy fitting method

## 1 Introduction

Thin-walled structures like cylindrical shells are widely used in civil engineering and aerospace shell structures. It is well known that this type of structure tends to buckle under axial compression, whereby this buckling occurs without prior notice. The buckling phenomenon is mostly accompanied by a sudden and severe drop in the bearing capacity of the cylindrical shell. In recent decades, lot of researchers have investigated shell stability from different viewpoints. However, the phenomenon of incomplete agreement of buckling load of cylindrical shells between theory and experiment under axial compression is a long-standing problem in shell structures [1]. Additionally, the buckling load results obtained from tests has a large dispersion. With the establishment of the nonlinear Donnell-Mushtari-Vlasov theory for large deflections of isotropic thin shells [2], pre-buckling consistency theory [3] and the initial post-buckling theory [4], combined with the nonlinear analysis based on finite element method (FEM), stability issues for cylindrical shells have been addressed at the theoretical level.

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However, the dispersion of experimental results and non-negligible errors between the theoretical analysis and test results still exist. The main reason for this is that it is difficult to obtain the exact geometric dimensions of the practical shell structure and accurate modelling. Nowadays, the three-dimensional (3D) scanning technique has been used to directly obtain actual geometric dimensions of shell specimens.

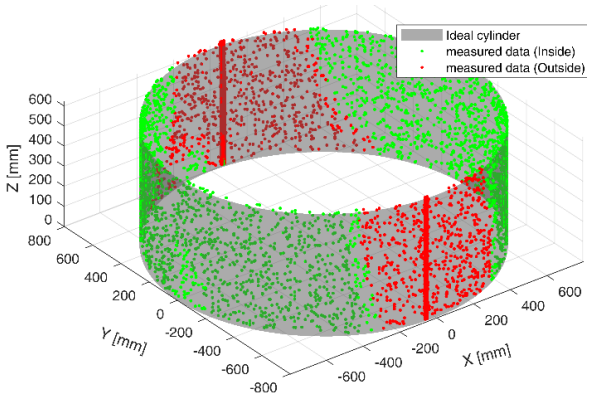



**Figure 1** Geometryy of shell specimen via 3D scanning, Source: [13]

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The lower bound curve is derived from a large number of scattered test data [5,6], which form the basis of design curves in current standards. At present, the "worst" geometric imperfection shape is commonly utilised to calculate the shell buckling load using numerical analyses and formulae. According to EN 1993-1-6 [7] and EN 1999-1-5 [8] standards, structural engineers need to introduce information on geometric imperfections into calculations to ensure the safety of shell structures. However, these design standards continue to have drawbacks; for example, the lack of determination of different types of imperfections, including misalignment, out-of-roundness, and radial deviation [9]. Theoretically, to meet structural reliability requirements and improve rationality and the accuracy of shell structures, the design of cylindrical shells should be based on statistical distribution values as opposed to lower bound values obtained by the "worst" imperfection [10].

Many attempts to resolve this conundrum through deterministic theory, experimentation, and probabilistic analysis have remained and numerous of them are not completely successful. Hence, this paper will focus on the analysis from the perspective of initial geometric imperfection, and use two methods to describe the randomness and spatial correlation of geometric imperfections. Then, the statistical samples of nonlinear buckling load of cylinder will be obtained based on obtained using the non-invasive stochastic finite element method (SFEM). After that the optimal probability function of cylinder bearing capacity will be calculated based on the maximum Entropy fitting method.

## 2 Representation of random geometric imperfection

A stochastic approach considering geometric imperfect for shell structures goes already back to the 1950s. One important step in the analysis of the influence of the imperfection properties randomness on bearing capacity of shell structure is the representation of the input of the imperfection uncertainty. Recently, there are two main ways to represent geometric imperfection which contain random properties, which treats the Fourier coefficients of the measured imperfection as random variables, namely random field and Fourier series representation with correlated random coefficients.

### 2.1 Random field and discretization

A random field can be seen as an extension of a random process in the spatial domain, which consists of an infinite number of random variables at points in the continuous domain. In general, the random field is a useful mathematical model to accurately describe the uncertain variables. With the continuous development of computer technology and finite element theory, the stability analysis of cylindrical shell is now commonly performed using commercial finite element software. Hence, the random field need to be discretized using a finite number of random variables and the purpose is to approximate the random field by the finite set of random variables. Thus, obtaining an approximate discrete random field with accurate statistical properties is the premise of studying the bearing capacity of shell structures with uncertain geometric imperfections. The discretization of random field can be simulated by using the Karhunen-Loève (K-L) series expansion [11] with truncated expansion after a finite M terms.

$$\hat{H}(\boldsymbol{X},\theta) = \mu(\boldsymbol{X}) + \sum_{i=1}^{M}\sqrt{\hat{\lambda}_i}\xi_i(\theta)\hat{\varphi}_i(\boldsymbol{X})\,. \tag{1}$$

where $\hat{H}(\boldsymbol{X},\theta)$ means a truncated random field in probability space; $\boldsymbol{X}$ is the position vector defined over the space domain and $\theta$ is primitive randomness; $\mu(\boldsymbol{X})$ is the mean function of the random field; $\xi_i(\theta)$ means a statistically uncorrelated random variable with zero mean; $\lambda_i$ and $\varphi_i(\boldsymbol{X})$ are the eigenvalue and the eigenfunction of the covariance kernel $C_{HH}$, which are the solution to the homogeneous Fredholm integral equation of second kind [88]. In general, the covariance matrix of random field can be approximated by the autocorrelation function $\rho_{HH}(\boldsymbol{X}_1,\boldsymbol{X}_2)$, i.e.:

$$C_{HH}(\boldsymbol{X}_1,\boldsymbol{X}_2) = \sigma(\boldsymbol{X}_1)\sigma(\boldsymbol{X}_2)\rho_{HH}(\boldsymbol{X}_1,\boldsymbol{X}_2) \tag{2}$$

where $\sigma$ means the standard deviation of random field. According to reference [12], isotropic exponential, squared exponential and sine functions can be employed as autocorrelation coefficient function for numerical analysis in most multidimensional homogenous and isotropic random fields. In this paper, the isotropic exponential function as shown in Eq. (3) are used for the random field of geometric imperfections.

$$\rho_{HH,1}(\boldsymbol{X}_1,\boldsymbol{X}_2) = \exp\left(-\frac{|\boldsymbol{X}_1-\boldsymbol{X}_2|}{L_c}\right) \tag{3}$$

Since discrete random field is approximated by finite truncation terms, this means that this approximate calculation ignores the high order terms in the tail and the random field will inevitably lose some uncertainty information. Therefore, the accuracy of the discrete random fields need to be evaluated. In general, the difference between the variance of the discrete random field and the target variance, as shown in Eq. (4), can be used as a measure of the accuracy of the truncated series.

$$\epsilon_{\sigma^2}(\boldsymbol{X}) = \frac{Var[H(\boldsymbol{X},\theta)]-Var[\hat{H}(\boldsymbol{X},\theta)]}{Var[H(\boldsymbol{X},\theta)]} \tag{4}$$

Figure 2 displayed the relationship between truncated terms $M$ and the standard deviation of discrete random field as well as the error measure for the discretization $\epsilon_{\sigma^2}(\boldsymbol{X})$. It is obvious that the discretization error decreases monotonically with the truncation order and the rate of decrease is constantly decreasing. This means that the increased computational effort after reaching a certain truncated terms $M$ cannot significantly improve the accuracy of discrete random fields. In this paper, the truncated terms $M$ is predefined as 2000 considering the balance between accuracy and consumed computing resources.

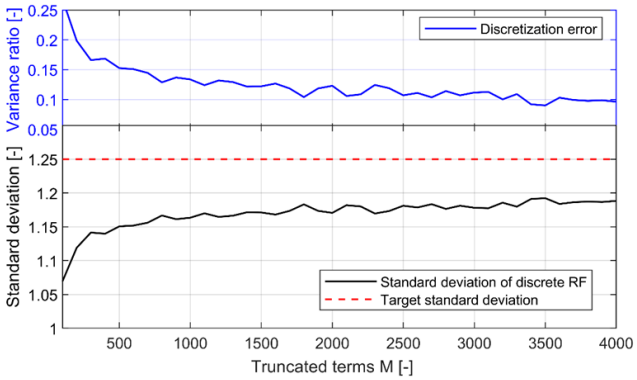



**Figure 2** Relationship between discretization and truncated terms. Source: Z. Li

Besides, the accuracy of discrete random field is also affected by the type of statistical distribution of the random field, especially random fields with high skewness. Although an arbitrary unknown distribution is nearly or approach to Gaussian distribution in most cases, because of the central limit theorem. The small probability events in the random field of geometric imperfection, which are sensitive to the tail regions of probability distributions, can be sensitive to the buckling bearing capacity of shell structures. Theoretically, the non-Gaussian random field with small skewness can be transformed using a translations processes or some mapping method based on the preceding discretization method of random field.

The establishment of a random field requires statistical distribution properties and spatial correlations, which can be described by the probability density function of random field variables and correlation lengths. The correlation length determines the decay of the mutual influence of two different points in random field and is a measure for a random field which are required to describe the random field with a given correlation property. Therefore, a correlation length equal to infinity describes a completely correlated random field and a correlation length tending towards zero describes an uncorrelated random field, which means that all random variables representing the random field are fully in dependent. In general, the radial deviations comparing perfect cylinder can be defined as geometric imperfections in cylindrical shell structures. According to the references [13], the measured geometric imperfections are composed of a non-uniform random field. This implies that the correlation structure of the geometric imperfections can be decomposed into two one-dimensional issues along the axial and circumferential direction and the correlation structure can be described with two separate correlation lengths. According to the fact that the defects in the circumferential direction of the cylindrical shell are dominated by low frequencies, Schenk et. al. [14] believe that the correlation lengths $L_{c,axial}$ and $L_{c,circum}$ can be approximated using the following equations.

$$L_{c,axial} = 0.6 \cdot H \tag{5}$$

$$L_{c,circum} = 0.06 \cdot 2\pi r \tag{6}$$

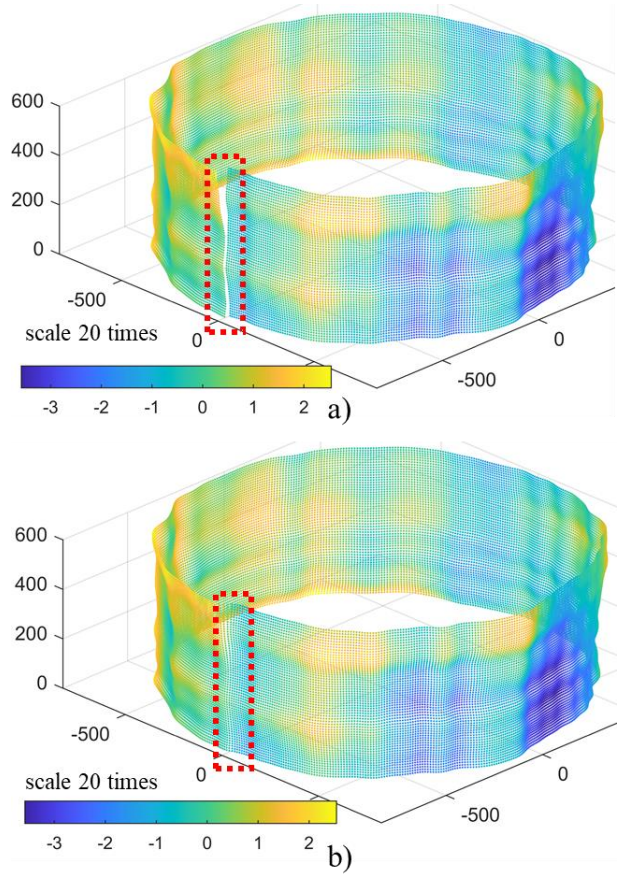



**Figure 3** Samples of random field for geometric Imperfections a) original and b) modified imperfection. Source: Z. Li

Furthermore, the discrete random fields are generally bounded, while a random field of cylindrical shell is unbounded in the circumferential direction. Therefore, the connection between the head and tail of the discrete random field in the circumferential direction needs to be corrected to make it transition smoothly and reduce singularities in numerical calculations. In this paper, the inverse distance weighted interpolation method [15] was employed to modify the imperfection values in the connection position. Figure 3 presents the a) original and b) modified imperfection random field.

### 2.2 Fourier series representation

Numerous studies shown that the actual geometry of the cylinder along the circumferential direction usually appears to be periodic, so the Fourier series method [16] can be used to analyze the geometric imperfection of shell structures. Usually, the geometric imperfections consist of a series of axial and circumferential waves based on Fourier series with a large number of sine and cosine functions. The radial deviation $w(h,\theta)$ ) of imperfection can be defined as a Fourier representation with the half wave cosine approach, as shown in Eq. (7).

$$w(h,\theta) = t * \sum_{k=0}^{N_k} \sum_{l=0}^{N_l} cos\left(\frac{k\pi h}{H}\right) * \left(A_{kl} cos\left(\frac{l\theta}{R}\right) + B_{kl} sin\left(\frac{l\theta}{R}\right)\right) \tag{7}$$

where $h$ and $\theta$ are the coordinates in the axial directions and circumferential phase shift angle, respectively. $H$ and $R$ mean the height and radius of the ideal circular cylindrical shell. $A_{kl}$ and $B_{kl}$ are the coefficients for the Fourier series of the half-wave cosine approach.

The coefficients $A_{kl}$ and $B_{kl}$ can be regarded as random parameters within the probabilistic analyses. However, these parameters are strongly correlated. Therefore, it is necessary to construct a coefficient vector and map the corresponding correlation. According to Eq. (8), the random vector $\boldsymbol{C}$ is composed of the Fourier coefficients $A_{kl}$ and $B_{kl}$.

$$\boldsymbol{C} = [A_{kl}, B_{kl}] \tag{8}$$

In general, the relation between a correlated random vector $\boldsymbol{C}$ and a random vector $\boldsymbol{\xi}$ with mean value 0 and variance 1 without dependence can be described by the Mahalanobis transformation [17] given by Eq. (9).

$$\boldsymbol{C} = \boldsymbol{\Sigma}^{\frac{1}{2}} \cdot \boldsymbol{\xi} + \boldsymbol{M} \tag{9}$$

where $\boldsymbol{\Sigma}$ is covariance matrix and $\boldsymbol{M}$ means mean value vector $\boldsymbol{C}$. Since the covariance matrix is a symmetric matrix, the roots of matrix $\boldsymbol{\Sigma}$ can be determined by the singular value decomposition, as shown in Eq. (10).

$$\boldsymbol{\Sigma}^{\frac{1}{2}} = \boldsymbol{U}\boldsymbol{\Lambda}^{\frac{1}{2}}\boldsymbol{V}^{*} \tag{10}$$

where $\boldsymbol{U}$ and $\boldsymbol{V}$ are the unitary matrix with eigenvectors of $\boldsymbol{\Sigma}$ and $\boldsymbol{\Lambda}$ is a diagonal matrix with the eigenvalues of $\boldsymbol{\Sigma}$. The idea of transforming with matrix $\boldsymbol{\Sigma}^{\frac{1}{2}}$ is similar to performing an inverse principal component analysis [18]. From the perspective of matrix analysis, the uncorrelated random vector is first rotated according to matrix $\boldsymbol{V}^{*}$. Since all variables in the original vector $\boldsymbol{\xi}$ are uncorrelated, this rotation will not cause changes of random vector. In some reference, this operation is directly omitted [19]. In fact, the Eq. (9) becomes the K-L series expansion. The main transformation is to stretch uncorrelated random vectors by $\boldsymbol{\Lambda}^{\frac{1}{2}}$ in all orthogonal directions, whereby the scale of stretch equal to the standard deviation of each random variables. Then the random vector will be rotated according to unitary matrix $\boldsymbol{U}$ based on the inverse principal components.

Therefore, under the premise of known covariance matrix and mean vector of coefficient vector for the Fourier series, coefficient vector samples can be randomly generated using this transformation based on the primitive random vector $\boldsymbol{\xi}$ and mean value vector of Fourier coefficient $\boldsymbol{M}$. The covariance matrix can be constructed using a similar method to those applied in random field theory based on the correlation matrix $\rho(c_i, c_j)$.

$$\boldsymbol{\Sigma} = \begin{pmatrix} Cov(c_1, c_1) & \cdots & Cov(c_n, c_1) \\ \vdots & \ddots & \vdots \\ Cov(c_1, c_n) & \cdots & Cov(c_n, c_n) \end{pmatrix} \tag{11}$$

$$Cov(c_i, c_j) = \sigma_i \sigma_j \cdot \rho(c_i, c_j) = {c_v}^2 \cdot \mu_{c_i} \mu_{c_j} \cdot \rho(c_i, c_j) \tag{12}$$

$$\rho(c_i, c_j) = exp\left(-\frac{\left(x_{c_i} - x_{c_j}\right)^2}{L}\right) \tag{13}$$

where $c_v$ means $L$ the coefficient of variation and dimensionless correlation length of the Fourier series coefficients. $X_{c_i}$ is the position vector for coefficient $c_i$ and $L$ can be seen as the correlation length of Fourier coefficient $A_{kl}$ and $B_{kl}$.

According to experimental outcomes, the geometric imperfections of cylindrical shells are basically dominated by a low frequency. Hence, the distribution of the Fourier coefficients is regular and can be fitted by a two-dimensional surface according to Eq. (15).

$$\boldsymbol{M} = \begin{bmatrix} \mu_{c_1} \\ \vdots \\ \mu_{c_n} \end{bmatrix} = \begin{bmatrix} \mu_{A_{kl}} \\ \mu_{B_{kl}} \end{bmatrix} \tag{14}$$

$$\mu_{A_{kl}} = A \cdot \exp\left(\frac{(k-K)^n}{N}\right) \exp\left(\frac{(l-L)^n}{N}\right) \tag{15}$$

where $A$ means the largest average of all coefficients. $K$ and $L$ mean the positions of the largest coefficients in the axial and circumferential directions, respectively. $N$ and $n$ are the shape coefficients of the 2D surface function, which can be determined by experimental investigation.

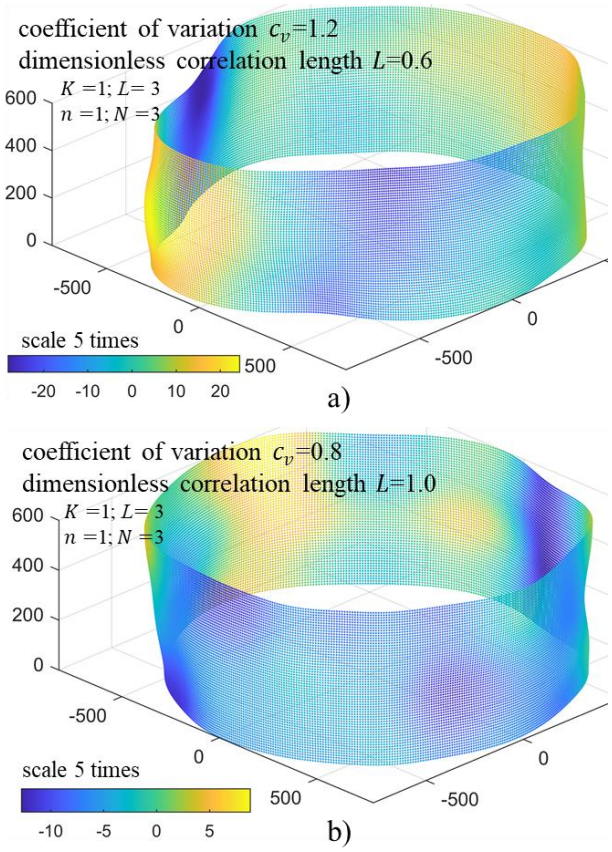



**Figure 4** Samples of Fourier series representation with various parameters. Source: Z. Li

Generally, for the considered set of shells, the Fourier series describes the imperfection patterns sufficiently accurate for $N_k$=10 and $N_l$=20 and hence, $2 \cdot (N_k + 1) \cdot (N_l + 1) = 462 \cdot$coefficients for $A_{kl}$ and $B_{kl}$ are need to be analysed. Therefore, the geometric imperfections with spatial correlation can be described by constructing a covariance matrix $\boldsymbol{\Sigma}$ based on coefficient of variation $c_v$ and the correlation length of Fourier coefficient $L$ as well as mean value vector $\boldsymbol{M}$ of Fourier coefficients. Figure 4 displays the geometric imperfection samples generated using Fourier series representation with different input parameters.

## 3 Probabilistic Buckling analysis of cylindrical shells and corresponding statistical distribution

### 3.1 Numerical calculations of buckling load

In this paper, numerical method with arc-length approach based on Geometrically and Materially Nonlinear Analysis with Imperfections included (GMNIA) is employed used to obtain the ultimate bearing capacity of cylindrical shells with random geometric imperfection based on random field and Fourier series representation. The non-intrusive Monte Carlo random finite element has high accuracy and robustness for the strong geometric nonlinear analysis, so it is very suitable for obtaining statistical samples of buckling bearing capacity of cylindrical shells. In order to make the numerical analysis and the random samples of geometric imperfection to interact with each other, the meshing rule of discrete random field or Fourier series representation and FE model are completely separated. The coordinates of the nodes in the finite element model will be modified through a somewhat mapping relationship, such as inverse distance weighted interpolation method [33], to reflect the randomly generated geometric imperfection. In this paper, the generated geometric imperfection files with random properties and finite element model node information are firstly read, and then the mapping relationship between each numerical point and the associated imperfection value points is established according to Eq. (1) in reference [15].

The numerical calculation of buckling bearing capacity is carried out using the Finite Element code ABAQUS®. The shell element S4R with 4 nodes and reduced integration points is used and all element size is defined as 10 mm. The all degrees of freedom for the node on the upper and lower edges of the numerical model are coupled with two reference points at the center of the upper and lower edges. This both reference points are defined as hinged. The elastic modulus and yield stress of the steel are defined as 210 GPa and 355 MPa, respectively.

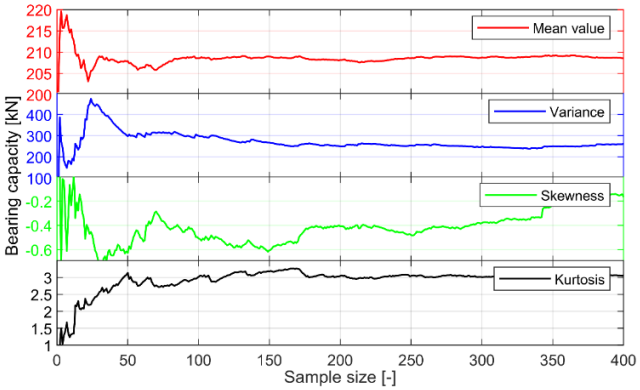



**Figure 5** Sample size of random field versus statistical moment of bearing capacity. Source: Z. Li

Figure 5 and 6 present the relationship of sample size of random field and Fourier series representation as well as the first four moments of the cylinder bearing capacity, namely mean value, variance, skewness and kurtosis, respectively. It is obviously that the first four moments show large fluctuations when the sample size is relative small and the fluctuation decreases as the sample size increases. When the sample size reaches 200, a relatively stable first four moments can be obtained,

which is very important for the subsequent statistical distribution function of bearing capacity and reliability analysis of shell structures.

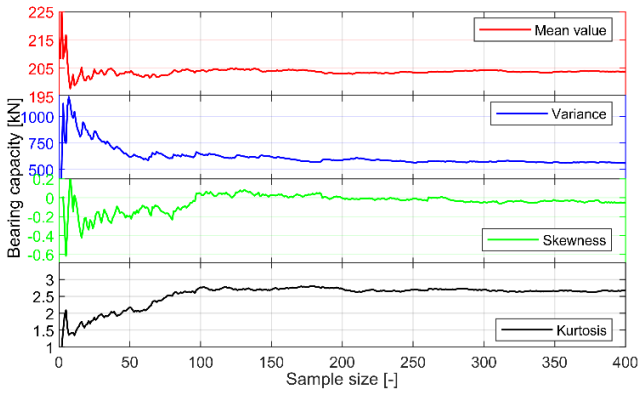



**Figure 6** Sample size of Fourier series representation versus statistical moment of bearing capacity. Source: Z. Li

### 3.2 Maximum Entropy fitting method

It is well known that the reliability analysis of structures is based on the statistical distribution functions of the action and resistance. In general, the Probability Density Function (PDF) of structural resistance can be approximated based on traditional fitting techniques. Usually, it is assumed that the sample belongs to some known random distribution function and then the null hypothesis theory is used to judge whether the data conform to the assumed distribution. If the distribution is checked for conformity, then the corresponding parameters of its distribution can be estimated. However, the accuracy of the results depends on the selected probability model and the sufficiency of the sample. Overall, the correct distribution of random variables and accurate corresponding statistical parameter are a prerequisite for reliability analysis. In some cases, where the sample size is small, the sample may accept some different distributions at the same time, and the statistical parameters of the random variable may be different from the true value.

Jaynes first proposed the principle of maximum entropy in [73]. Based on additional constraints under known information, he pointed out that the distribution with the smallest deviation has the largest entropy. Therefore, the maximum entropy principle provides a way to construct an optimal probability distribution under known information. In this paper, the maximum entropy approach is employed to fit the optimal probability density distribution for the buckling bearing capacity of shell structures based on calculated 200 samples. The ideal of the maximum entropy fitting approach is to find a probability density function $f(x)$ with the maximum entropy $H$ under the constraints [21]. This is formulated as a constrained optimization problem:

$$\begin{cases} \text{Maximize} & \mathrm{H} = -\int \mathrm{f(x)lnf(x)dx} \\ \text{subject to} & \mathrm{E}\{\phi_n(x)\} = \int \phi_n(x)\mathrm{f(x)dx} \end{cases} \tag{16}$$

where $\phi_0(x) = 1$ and $\phi_n(x) = x^n$. $E\{\phi_n(x)\}$ is the $n$th moment of the random variable. The conventional method to solve the optimization problem is to introduce the Lagrange multiplier $\lambda_n$. The $\boldsymbol{\lambda} = [\lambda_0 \cdots \lambda_n]$ is obtained by solving the set of $n+1$ nonlinear equations as follows:

$$\int \phi_n(x)\exp[-\textstyle\sum_{n=0}^{N}\lambda_n\phi_n(x)]dx = \mathrm{E}\{x^n\} \tag{17}$$

Therefore, the fitted PDF of buckling bearing capacity can be defined as a form containing Lagrange multipliers, as following,

$$f(x) = exp[-\textstyle\sum_{n=0}^{N}\lambda_n\phi_n(x)] \tag{18}$$

Theoretically, the collection of all the moments can uniquely determine the distribution. All moments mean that the value of $N$ tends to infinity, which is not achievable in numerical operations. Usually, the use of the first four moments in reliability analysis, such as fourth-order moment reliability method, has already a fairly high accuracy. In this paper, the optimal function of PDF is determined based on the first four moments.

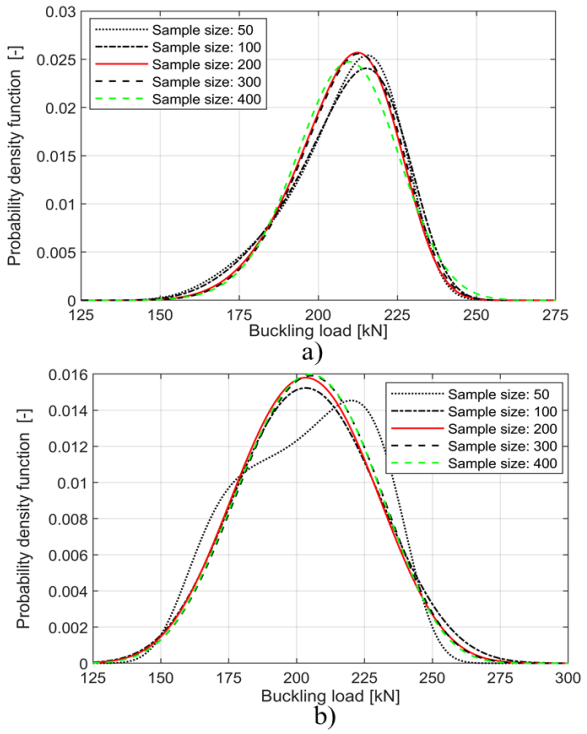



**Figure 7** Optimal fitted PDF of nonlinear buckling load a) random field model b) Fourier series representation model. Source: Z. Li

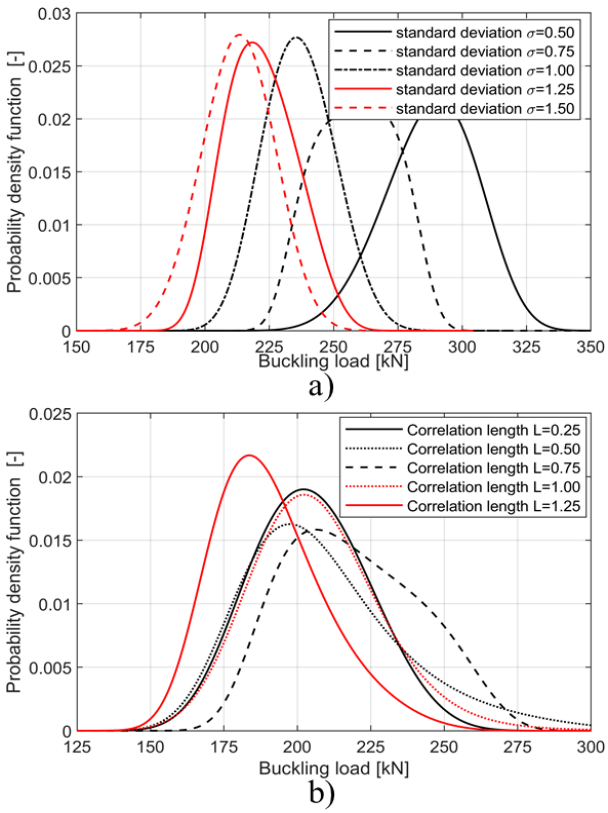



**Figure 8** Influence of different parameters, a) standard deviation of random field ( $L_{c,axial} = 0.6 \cdot H$, $L_{c,circum} = 0.06 \cdot 2\pi r$ ) and b) correlations length of Fourier series representation model ($c_v = 0.8$) on PDF of cylinder bearing capacity. Source: Z. Li

Figure 7 show the best fitted PDFs for random field and Fourier series

representation model using maximum Entropy fitting method based on the first four statistical moments with different sample size. Obviously, more samples at the expense of large computational resources can get more accurate PDFs. The outcomes show that when the sample size is greater than 200, the fitted PDF basically remains unchanged.

In general, all parameters in the random field and Fourier series representation model will affect the fitted PDF. For example, Figure 8 shows the optimal PDF obtained with different standard deviations for random field and various correlation lengths for Fourier series representation model. It is clearly that the mean value of ultimate bearing capacity of cylindrical shells decreases with the increase of random field standard deviation. Besides, there is a nonlinear relationship between the dimensionless correlation distance of the Fourier series coefficients and the average value of the cylinder nonlinear buckling load. The largest mean value of cylinder bearing capacity is obtained when the correlation coefficient takes about 0.75. Moreover, the obtained PDFs of buckling load may not conform to a normal distribution, although the primitive random variables distribution is Gaussian.

## 4 Conclusion and Outlook

In this paper, two models, i.e., random field and Fourier series representation model are discussed and developed to describe the initial geometric imperfections with randomness and spatial correlation for cylindrical shell structures. Non-Invasive stochastic finite element method is employed to simulated the nonlinear buckling load of cylindrical shells based on geometrically and materially nonlinear analysis with imperfections included approach. The first four statistical moments of the numerical bearing capacity samples are calculated and the optimal probability density function of buckling load is obtained using maximum Entropy fitting method based on the statistical moments. The results indicate that both models are able to quantify the statistical properties of the bearing capacity due to the randomness of geometric imperfections. As such, based on the proposed approach, stochastic properties can be directly considered in numerical calculations to develop a more precise and economical methodology for the design of steel structures.

However, the parameters in these models require further study to reveal their relationship to the statistical distribution function of the bearing capacity. Especially the model parameters need to be analyzed and obtained based on the measured geometric imperfection with a large number of experiments. Besides, the sensitivity analysis of the parameters of the both models to the bearing capacity of cylinder needs to be performed in the future.

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